

# Computer Science Department Examination

Subject: Operating Systems & Data Structures

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## Section A — Multiple Choice Questions

**1. What is the primary function of an operating system kernel?**

- (A) Word processing and spreadsheet calculations
- (B) Managing hardware resources and process execution
- (C) Sending and receiving external network packets only
- (D) Compiling web application assets

**2. Which data structure operates on a Last-In, First-Out (LIFO) discipline?**

- (A) Queue
- (B) Balanced Binary Search Tree
- (C) Stack
- (D) Circular Buffer

**3. What is the asymptotic time complexity of searching an element in a balanced BST?**

- (A)  $O(n)$
- (B)  $O(\log n)$
- (C)  $O(n \log n)$
- (D)  $O(1)$

## Section B — Short Answer Questions

**4. Explain the key distinctions between a process and a thread.**

**5. Describe the role of page tables and TLB in virtual memory management.**